

# Loan Default Risk Analysis

Business Insights & Predictive Modeling for Lending Portfolios

Executive Brief



# Project Overview



## Project Summary

This project focuses on analyzing loan default data to identify patterns, trends, and financial risk indicators within a lending portfolio. Using Python and data analytics, the dataset was cleaned, transformed, and explored through statistical analysis and visualization.



## Final Deliverable

An interactive dashboard built with Dash and Plotly that enables dynamic exploration of loan default trends and lending insights through real-time filtering.

### Technologies Used

Python, Pandas, Seaborn, Matplotlib, Plotly, Dash, Visual Studio Code



## Core Analysis Areas

- Data cleaning & preprocessing
- Exploratory data analysis (EDA)
- Statistical analysis
- Data visualization
- Predictive modeling
- Dashboard development

## Key Business Insights

- Regional loan default patterns
- Credit score distribution analysis
- Monthly loan amount trends
- Borrower financial indicators
- Relationships between lending variables and risk

# Analyzing Risk to Strengthen Lending Strategies

1

## The Objective

Analyze loan default data to identify patterns, financial risk indicators, and borrower behaviors within the lending portfolio.

2

## The Process

Data cleaned, exploratory data analysis conducted, and predictive models built to support data-driven underwriting.

3

## The Findings

Credit scores alone are insufficient; commercial loans carry higher default rates, and loan amounts correlate strongly with property values.

# Core Objectives of the Analytics Initiative



## Understand Behavior

Analyze borrower risk profiles and repayment habits to evaluate individual reliability.



## Identify Patterns

Detect regional, categorical, and financial default trends across the portfolio.



## Explore Relationships

Map connections between income, debt burdens, and total loan amounts.



## Build Models

Develop regression models to predict loan volumes and proactively assess risk.

# Ensuring Data Integrity for Reliable Analysis



# Default Rates Vary Significantly By Region & Purpose

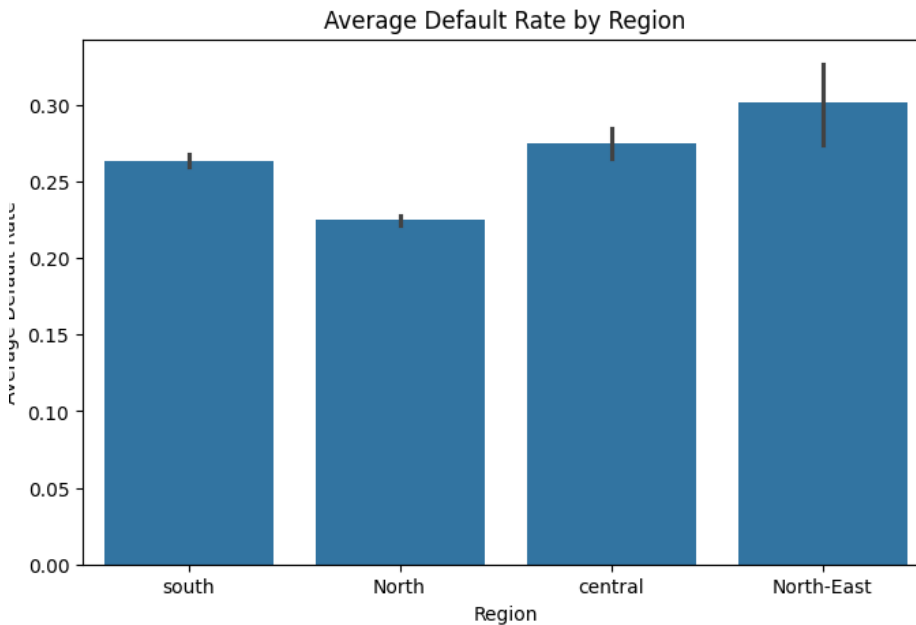
## Regional Disparities

Default rates fluctuate geographically. The North-East region shows the highest average default rate, highlighting the need for location-aware risk assessment.

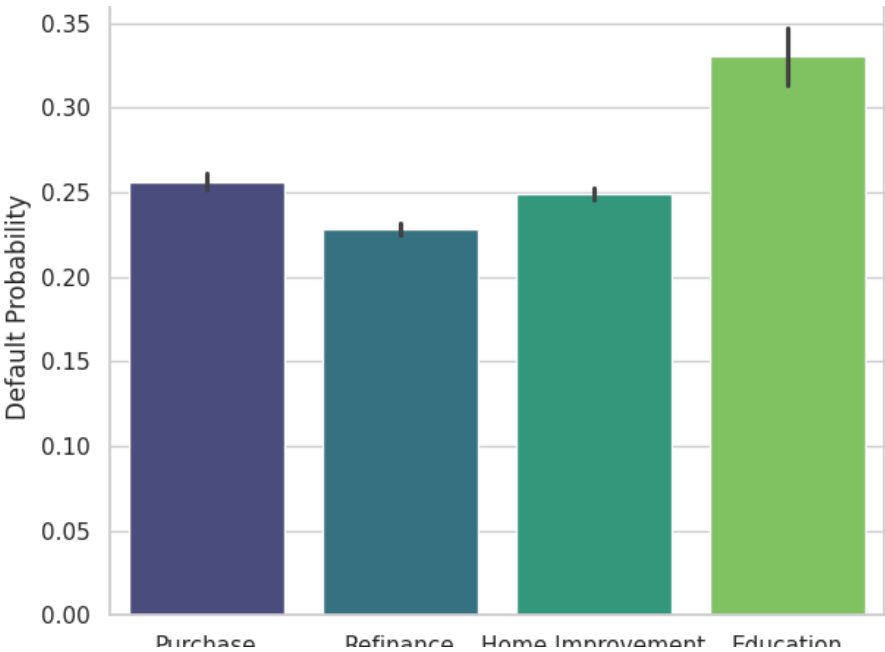
## Purpose-Driven Risk

Different loan purposes exhibit distinct default probabilities. Education and Purchase loans show higher default frequencies compared to Refinancing, requiring tailored underwriting strategies.

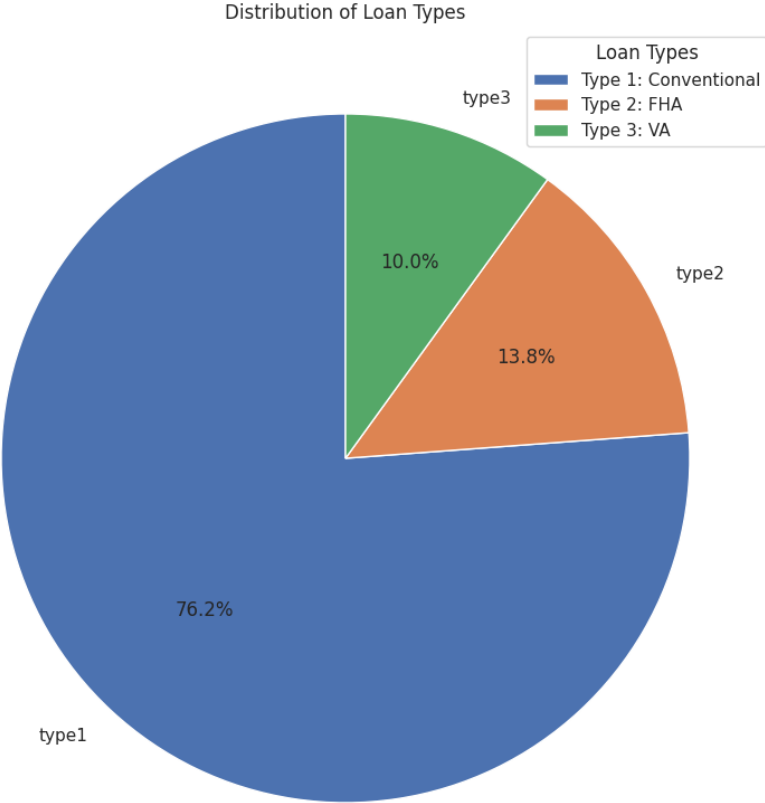
Average Default Rate by Region



Default Rate by Loan Purpose

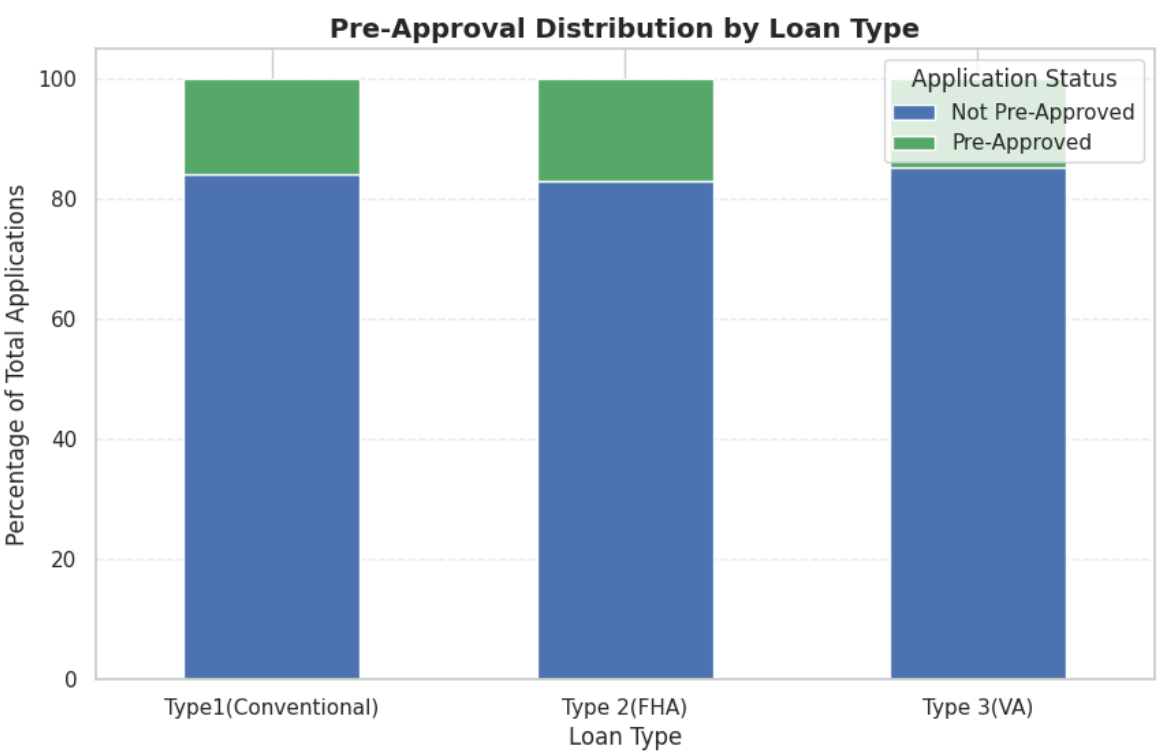


# Conventional Loans Dominate the Portfolio



## Portfolio Composition

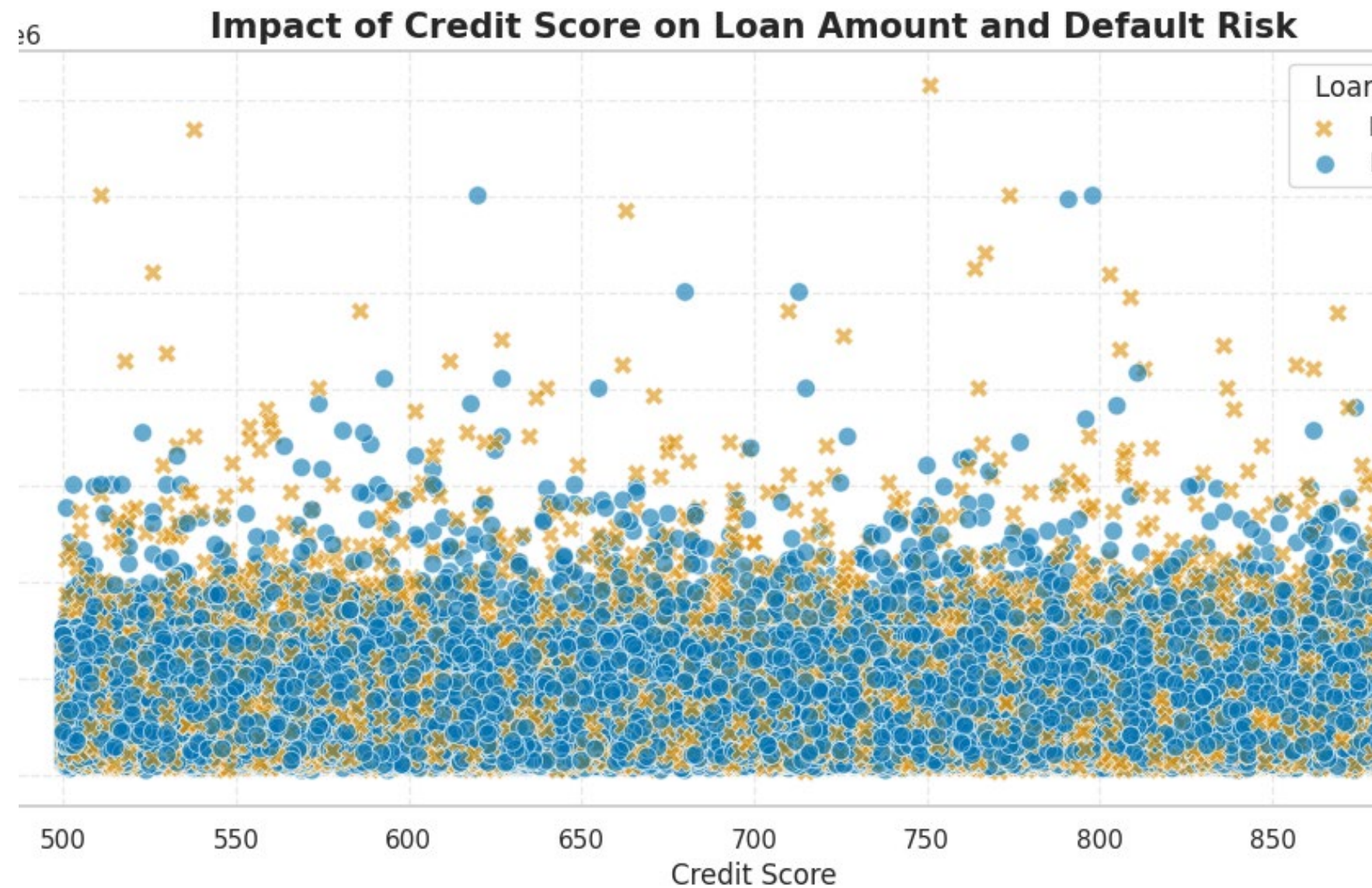
Conventional loans make up the vast majority (76.2%) of the dataset, driving the core risk profile of the lending portfolio.



## Pre-Approval Trends

The majority of applications across all loan categories were not pre-approved, suggesting reactive processing workflows rather than proactive screening.

# Credit Score Alone Does Not Fully Predict Defaults



## Score Distributions

Most borrowers fall into moderate-to-high ranges (median ~700). Stronger creditworthiness provides access to larger lending opportunities.

## Complex Default Behavior

While defaulted loans concentrate in lower tiers, significant overlap exists. This scatter plot visually proves that credit scores cannot act as the sole determinant for default prediction.

# Beyond Credit Scores: Uncovering Complex Risk Factors



## Commercial vs. Residential

Commercial loans show noticeably higher default rates despite having almost identical average credit scores (~700). Other operational risks clearly dominate here.



## Volume Correlations

Loan amounts demonstrate strong positive correlations with both borrower income and property value, indicating tight underwriting caps on lending limits.



## Interest Rate Dynamics

Interest rate spreads demonstrate negative relationships with larger loans and higher property values. Bigger loans secure better rates, but carry massive absolute risk.

# Building the Predictive Modeling Pipeline

1

## Baseline Models

Started with simple and multiple linear regressions to map basic financial relationships.

2

## Polynomial Features

Applied second-order transformations to capture complex, non-linear relationships.

3

## Ridge Regularization

Introduced penalty terms ( $\alpha=0.1$ ) to stabilize coefficients and prevent model overfitting.

4

## Train/Test Validation

Split data (15% test) to validate the  $R^2$  score and ensure robust generalization.

# Multi-Factor Evaluation is Critical for Risk Assessment

## Segment Vulnerability

Specific loan purposes and commercial lending segments carry elevated intrinsic risk, completely independent of the borrower's personal credit score.

## Portfolio Complexity

Lending risk is not isolated to single variables. Accurate prediction requires a holistic evaluation of income, property value, and rate spreads interacting simultaneously.

## Borrower Variability

Credit scores provide a useful baseline, but overlapping default patterns demand deeper financial profiling including income and debt burdens.

# Strategic Recommendations for the Lending Portfolio

1

## Enhance Underwriting

Shift from credit-score-dominant decisions to comprehensive, multi-variable risk scoring models.

2

## Targeted Assessment

Apply stricter review criteria and adjusted rate thresholds for commercial loans and higher-risk loan purposes.

3

## Proactive Monitoring

Track interest rate spreads and debt-to-income ratios systematically as leading indicators of portfolio stress.

4

## Deploy Analytics

Integrate regularized predictive models into the daily borrower evaluation and pricing workflow.

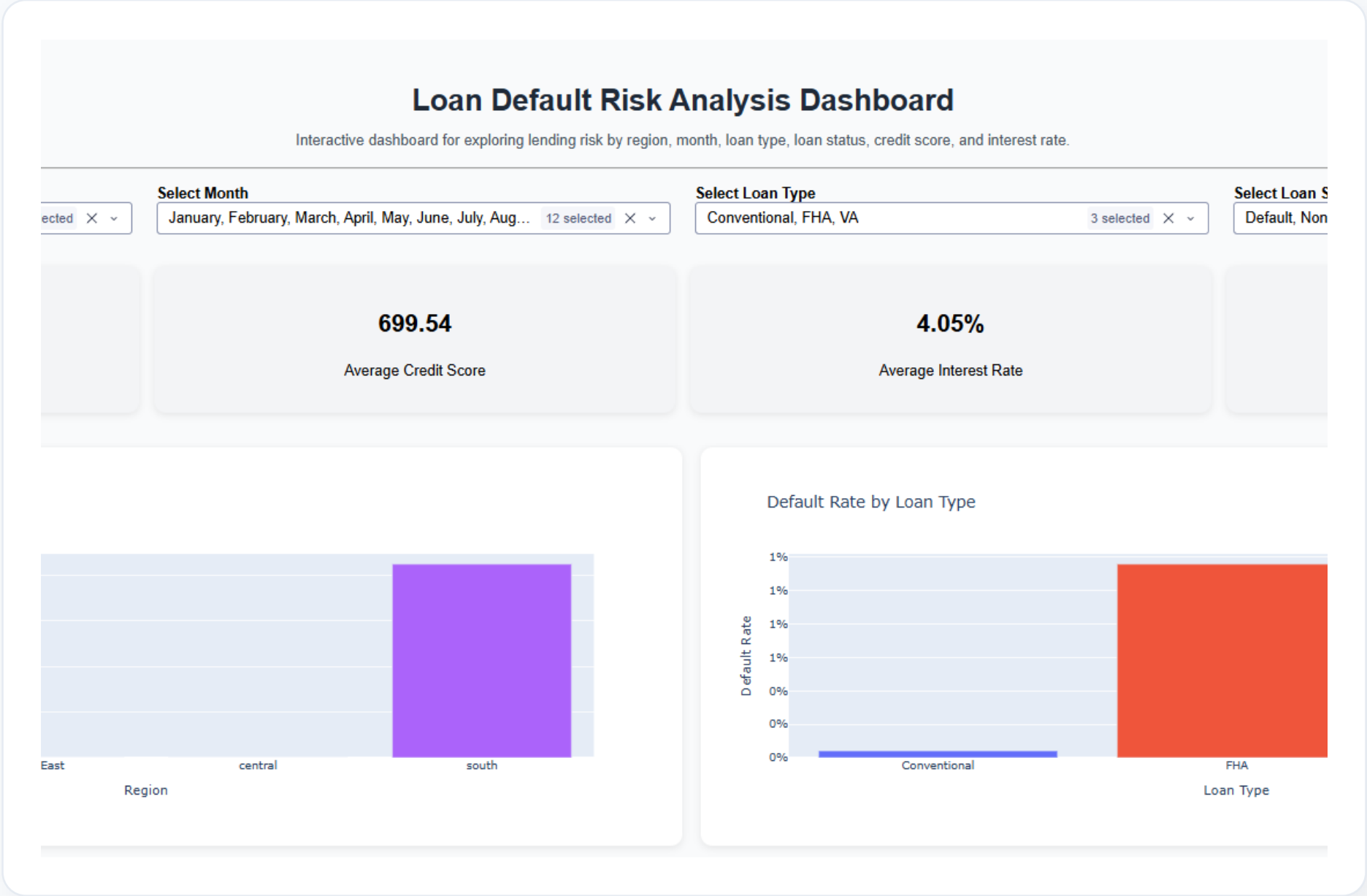
# Interactive Dashboard Overview

## Enhanced Usability

The environment is optimized for exploratory analysis via responsive dashboard design. We added powerful multi-filter interactivity for Region, Month (by name), Loan Type, and Loan Status to improve stakeholder comparative analysis.

## Key Capabilities

- Real-time KPI updates
- Dynamic chart filtering
- Plotly visualizations
- Advanced Dash callback logic



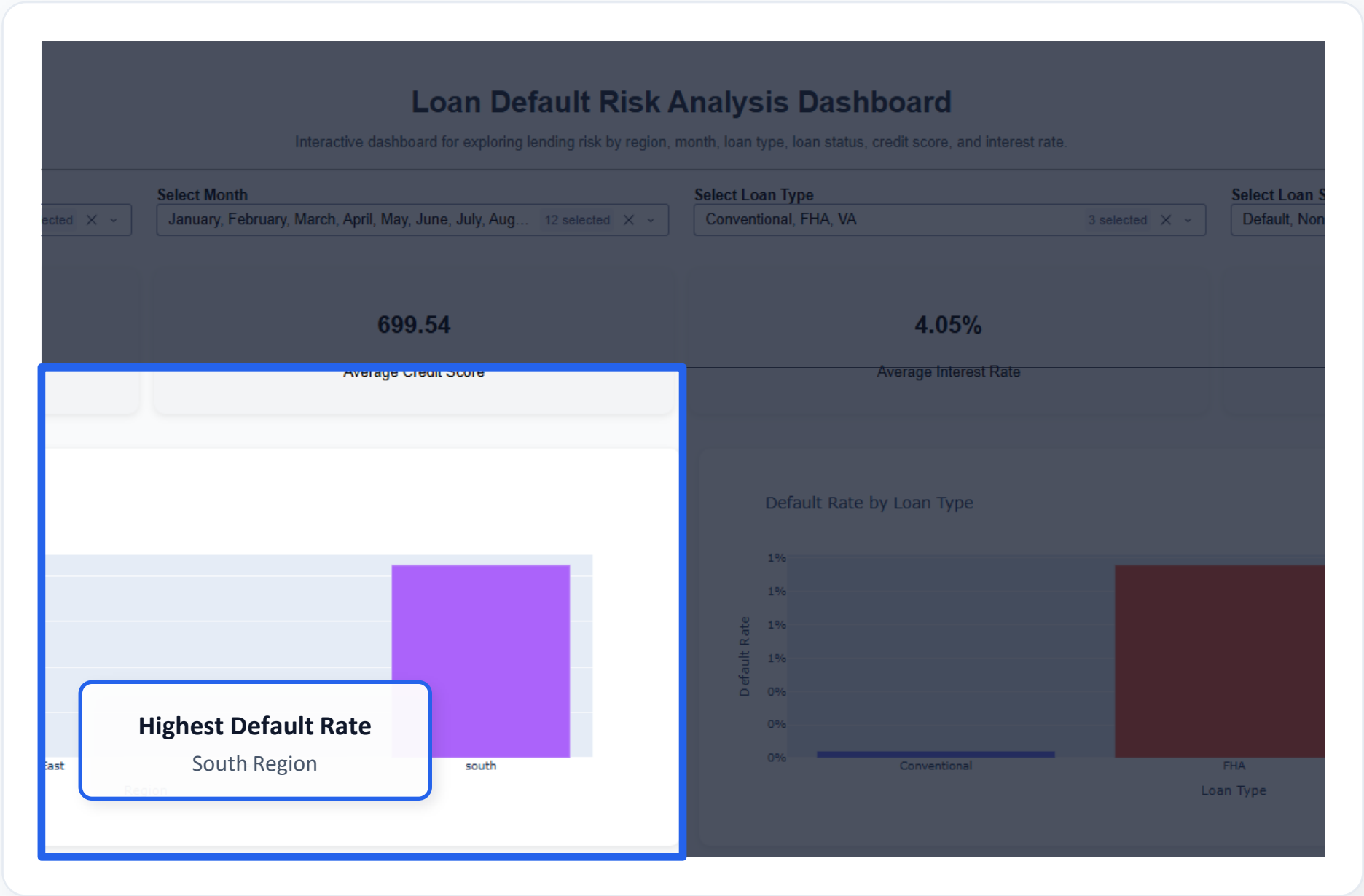
# Regional Default Risk Insights

## Geographic Vulnerability

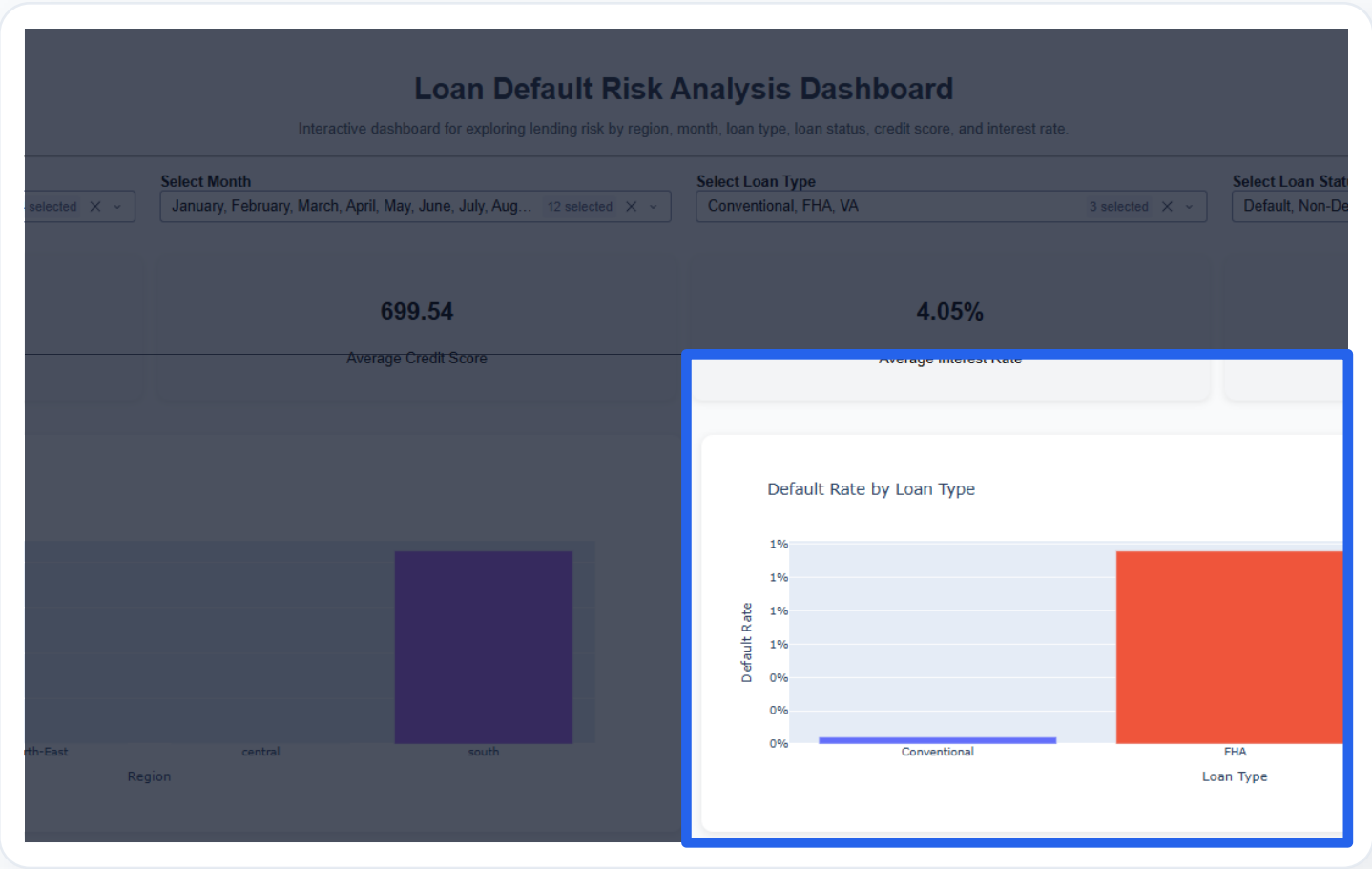
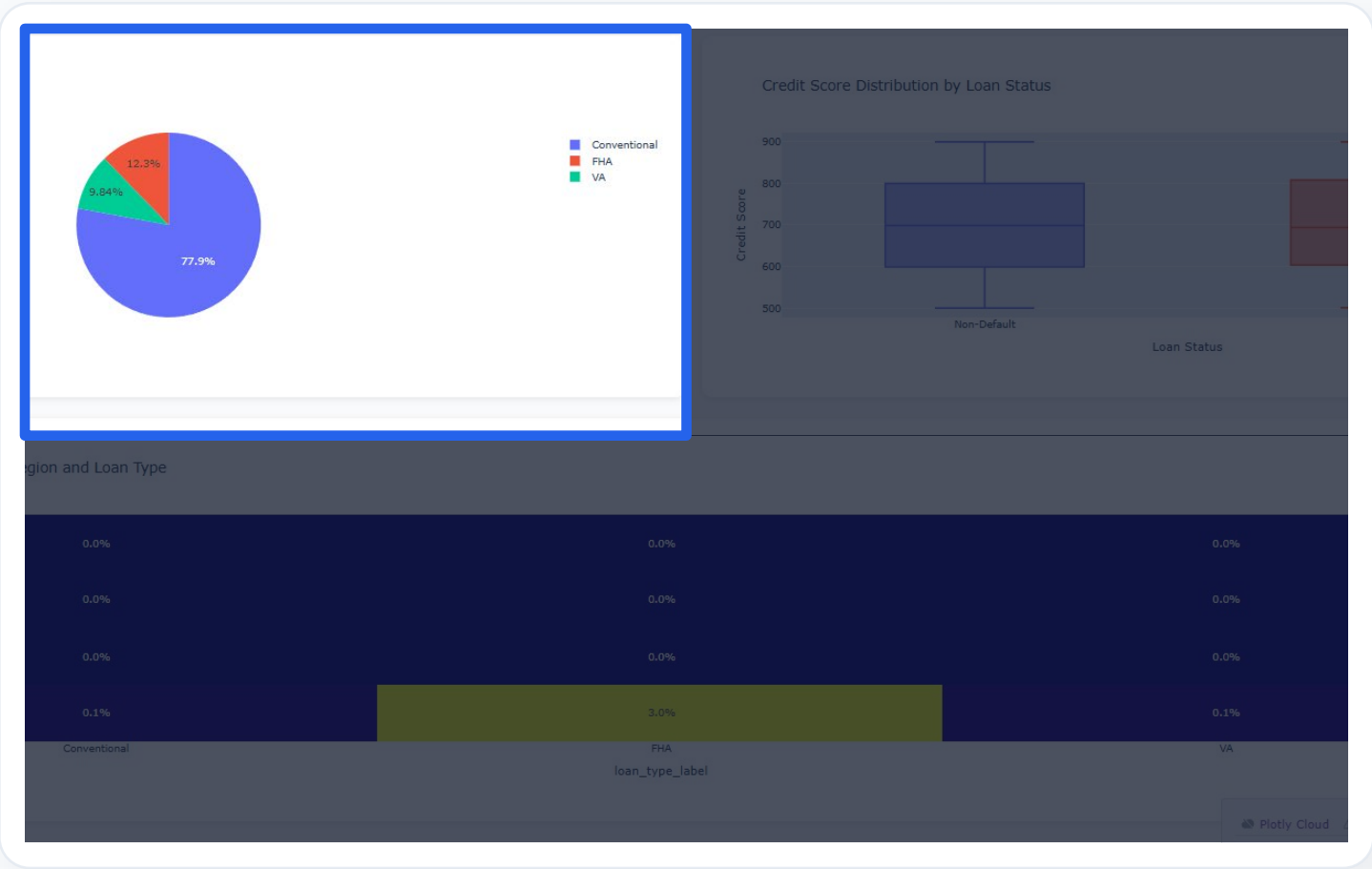
The dashboard reveals significant disparities across geographies, with the South region demonstrating the highest observed default activity.

## Strategic Action

This regional segmentation supports targeted risk mitigation strategies. Lending policies and underwriting thresholds can be adjusted locally to reflect these differing baseline risks, rather than applying a blanket national policy.



# Loan Type Risk Comparison



## The Accessibility vs. Risk Trade-Off

When comparing the two dashboard views, a clear divergence emerges: FHA loans represent a relatively small portion of the overall portfolio distribution (left), yet they demonstrate a substantially elevated default risk (right). This highlights the ongoing strategic tension between maximizing lending accessibility and managing portfolio risk.

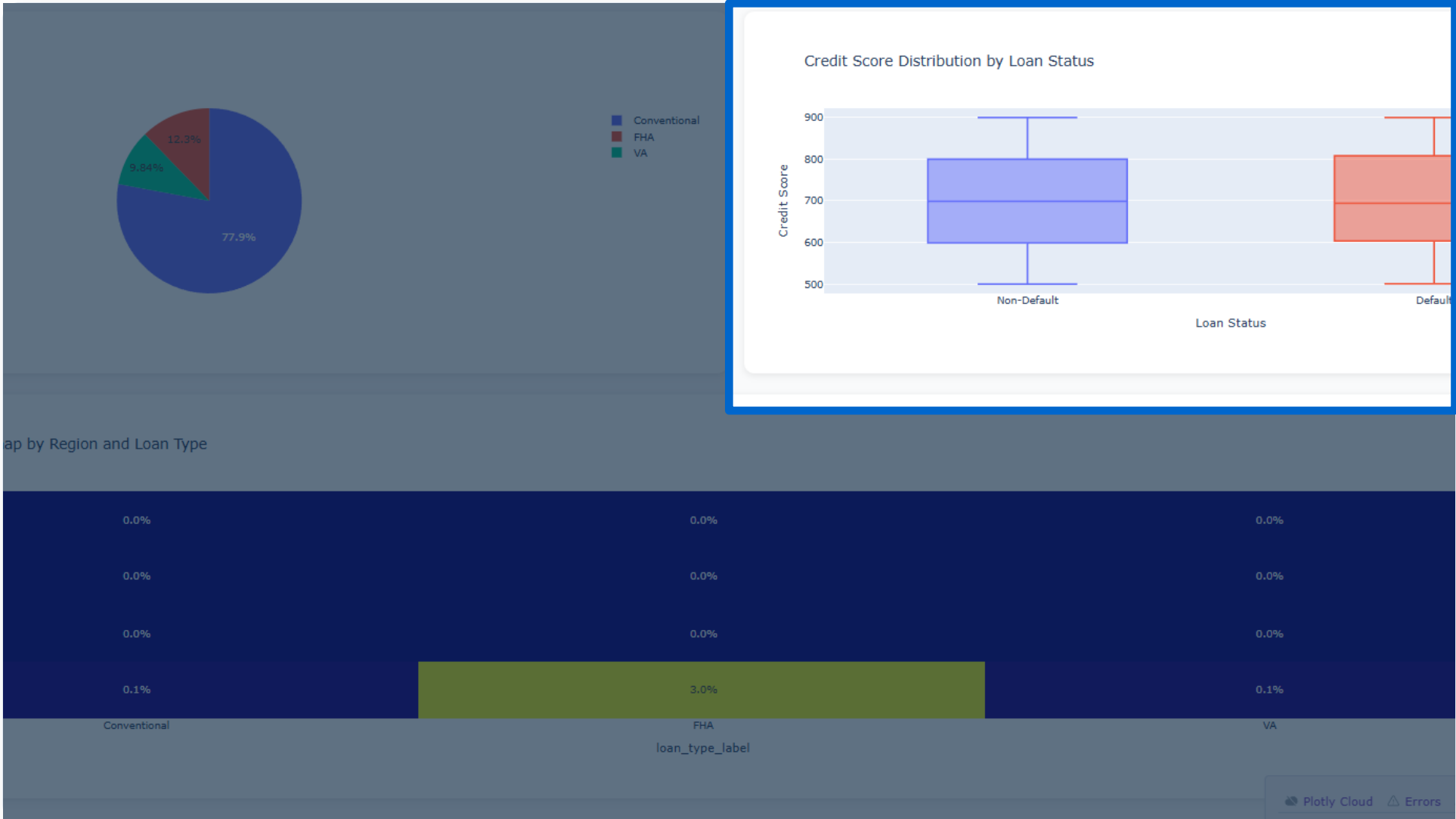
# Credit Score and Loan Performance

## The Credit Score Limitation

As shown in the dashboard's distribution analysis, credit scores alone are not strong, isolated predictors of default outcomes. Both defaulted and non-defaulted loans show similar credit score distributions.

## Multi-Factor Evaluation

This confirms the absolute necessity of multi-factor risk evaluation. Accurate underwriting must systematically incorporate debt-to-income ratio, property value, and interest rate spreads alongside the baseline credit score.



# Final Business Recommendations



## Regional Monitoring

Implement localized risk monitoring, applying adjusted underwriting policies to high-default regions like the South.



## Strengthen FHA

Strengthen risk assessment criteria specifically for FHA loans to balance accessibility with acceptable portfolio risk.



## Improve Modeling

Advance predictive modeling by integrating multiple financial indicators (DTI, property value) beyond credit scores.



## Expand Dashboards

Expand dashboard capabilities to include real-time portfolio monitoring and forward-looking default forecasting.

## Key Takeaways

Integrating real-time visualization with multi-factor predictive models transforms raw loan data into a proactive risk-management engine. By shifting focus from isolated metrics to holistic borrower profiles, we can safely expand lending accessibility while systematically suppressing default exposure.

# Data Analytics Drives Confident Financial Decisions

By transforming raw borrower data into actionable insights, lending institutions can optimize risk assessment and strengthen portfolio performance.

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